

Test Project

Mobile Applications Development

Module A

Introduction

With the continuous penetration of the Internet into the real economy and advances in technology, the main battlefield of retail banking business has gradually shifted from physical offline branches to mobile banking apps. Through bank card apps, users can track income and expenditure analyses in real time.

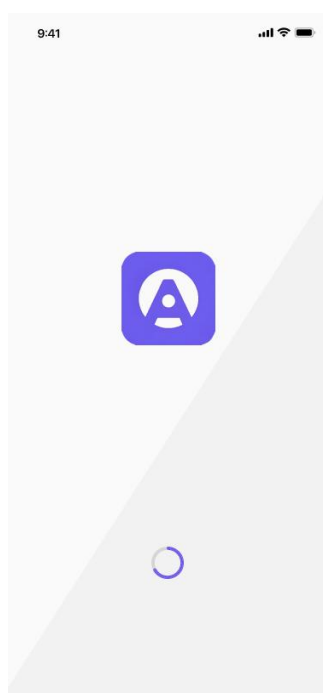
Description of project and tasks

This development needs to be carried out in strict accordance with the style of the prototype and the requirements. The task is divided into two parts according to different equipment types:

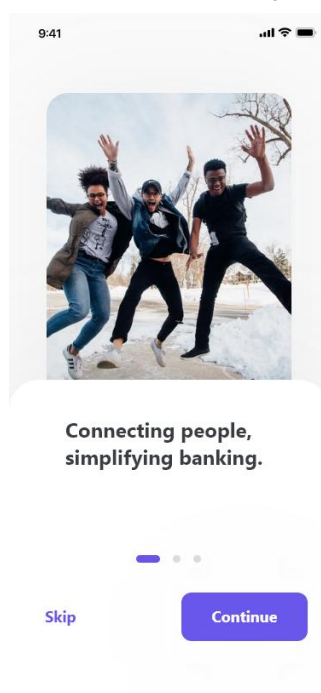
Name	Competition time	Marking devices
Module A	2.5	Emulator: Pixel 2 or iPhone 8 Plus

The following is an overview of some prototypes. Please refer to the **media-files** for details .

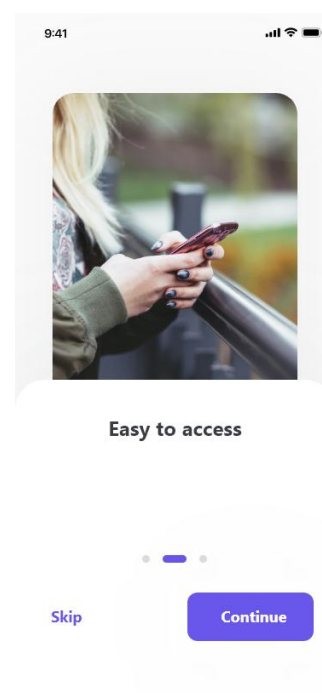
Application prototypes



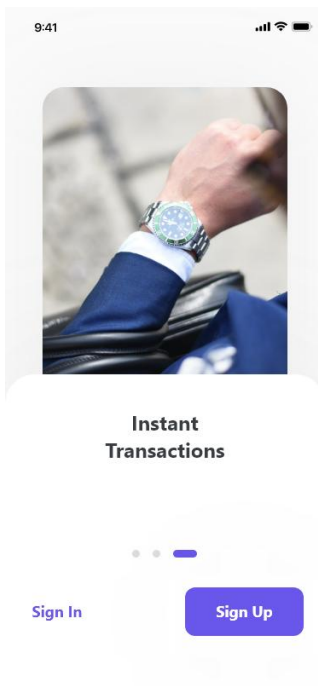
Splash Screen



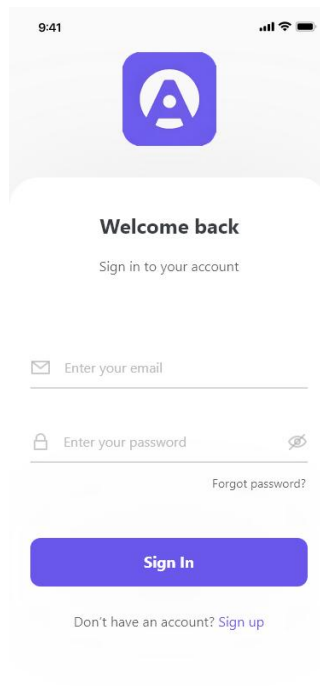
Onboarding Page-1



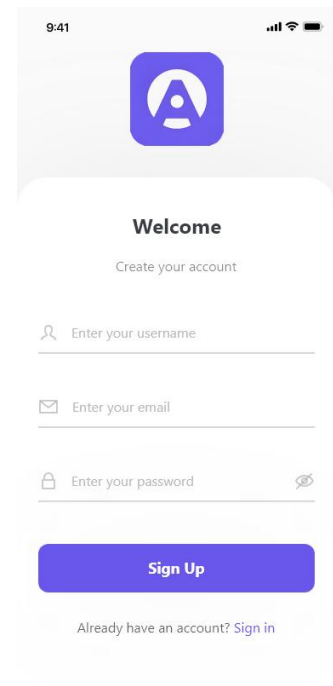
Onboarding Page-2



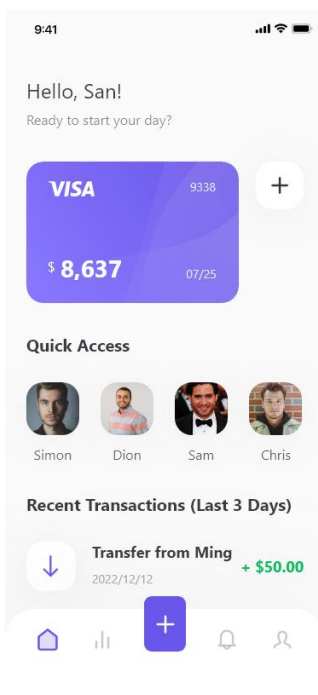
Onboarding Page-3



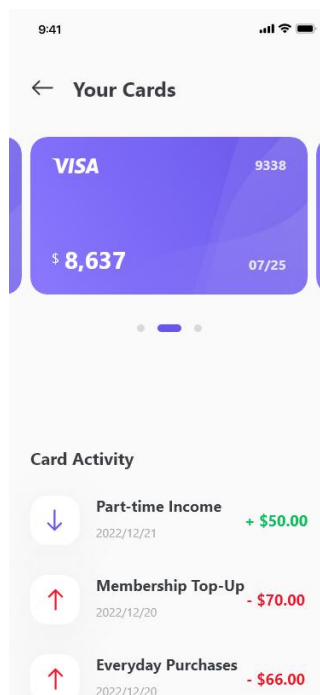
Sign In Page



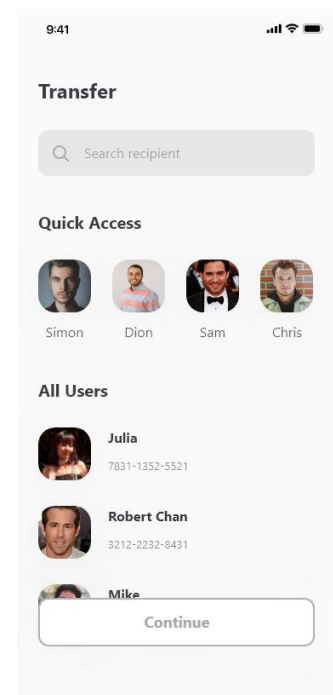
Sign Up Page



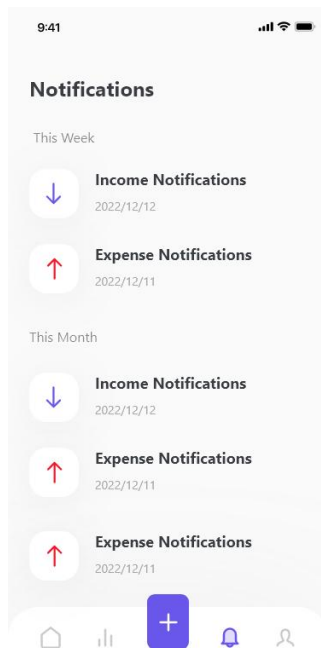
Home Page



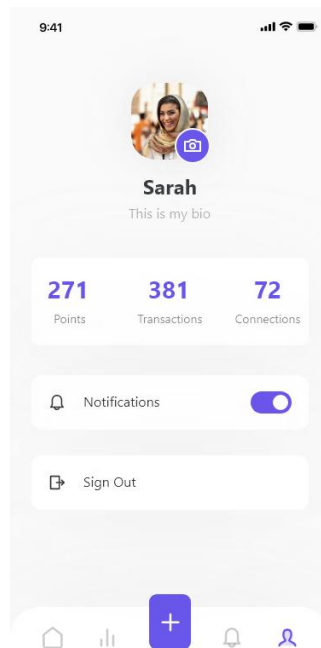
Card Wallet Page



Transfer Page



Notifications Page



Profile Page

General Demands

1. Strictly follow the details of the prototypes.

Pages Demands

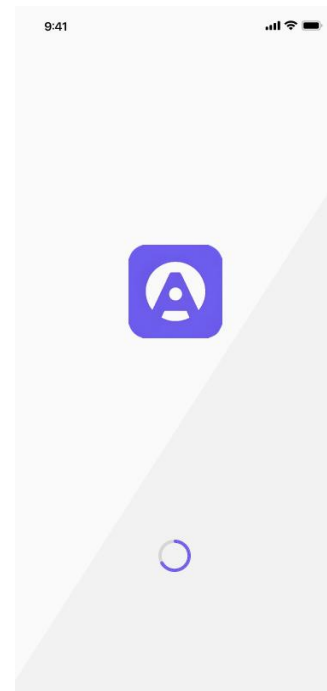
1. Splash Screen

1.1 Elements include:

- 1) Logo;
- 2) Circular progress bar.

1.2 Functional requirements :

- 1) When the app is opened, the splash page is displayed;
- 2) The circular progress bar continues rotating for three seconds;
- 3) After waiting for three seconds, the app automatically jumps to the onboarding page.



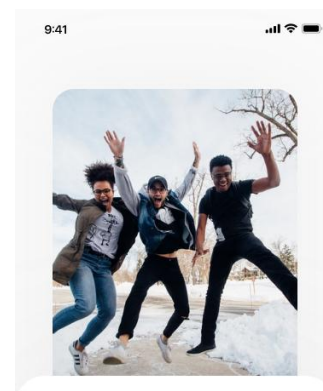
2. Onboarding Page

2.1 Elements include:

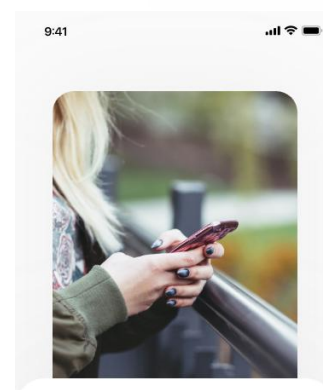
- 1) Image;
- 2) Introductory text;
- 3) **【Skip】** button;
- 4) **【Continue】** button;
- 5) The third page includes an image, introductory text, **【Sign In】** button, and **【Sign Up】** button;
- 6) Page indicator dots.

2.2 Functional requirements:

- 1) The onboarding flow consists of three pages;
- 2) Tapping the **【Continue】** button navigates to the next page; tapping the **【Skip】** button jumps to the third page;
- 3) On the third page, tapping **【Sign In】** jumps to the login page; tapping **【Sign Up】** jumps to the registration page;
- 4) All three pages have page indicator dots; the selected page is indicated by a capsule-shaped dot, while unselected pages are indicated by round dots;
- 5) Users can swipe left or right to switch between the three pages.

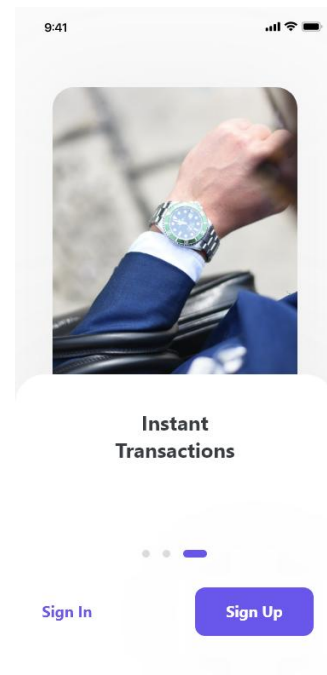


Connecting people,
simplifying banking.



Easy to access





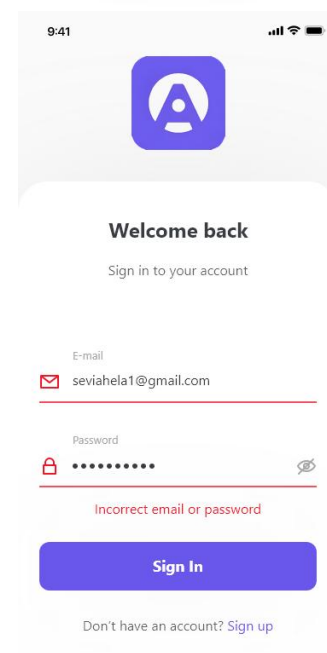
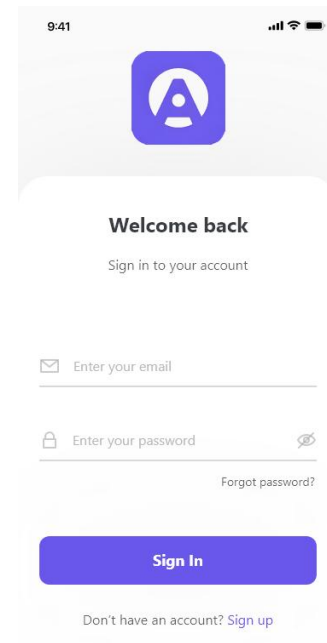
3. Sign In Page

3.1 Elements include:

- 1) Logo;
- 2) **【Welcome back】** text;
- 3) **【Sign in to your account】** text;
- 4) Email input field;
- 5) Password input field;
- 6) Show/Hide Password button;
- 7) **【Forgot password?】** button;
- 8) **【Sign In】** button;
- 9) **【Don' t have an account? Sign up】** button.

3.2 Functional requirements:

- 1) The email and password can only be entered in a single line; line breaks are not allowed;
- 2) The password is hidden (obscured) by default; tapping the **【Show/Hide Password】** button toggles its visibility;
- 3) Tapping **【Don' t have an account? Sign up】** navigates to the registration page;
- 4) If the email or password field is empty when **【Sign In】** is tapped, display the prompt: **【Incorrect email or password】** ;
- 5) If both the email and password fields are filled when **【Sign In】** is tapped, the button text is replaced with a rotating circular progress bar (rotates continuously for three seconds). After three seconds, the app automatically jumps to the home page.



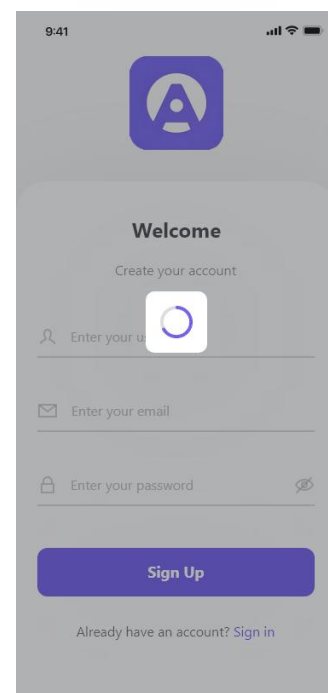
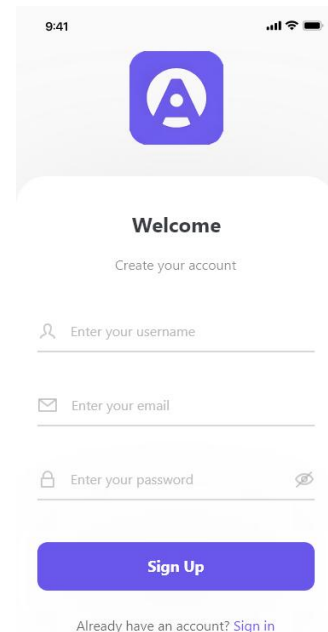
4. Sign Up Page

4.1 Elements include:

- 1) logo;
- 2) **【Welcome】** text;
- 3) **【Create your account】** text;
- 4) Username input field;
- 5) Email input field;
- 6) Password input field;
- 7) **【Sign Up】** button;
- 8) **【Already have an account? Sign in】** button.

4.2 Functional requirements:

- 1) Tapping the sign-up button displays a progress dialog. After three seconds, the app automatically navigates to the home page.



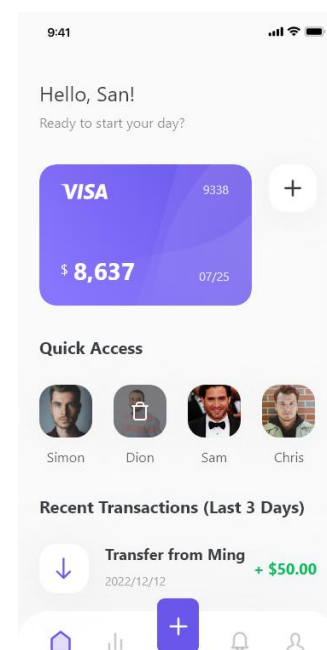
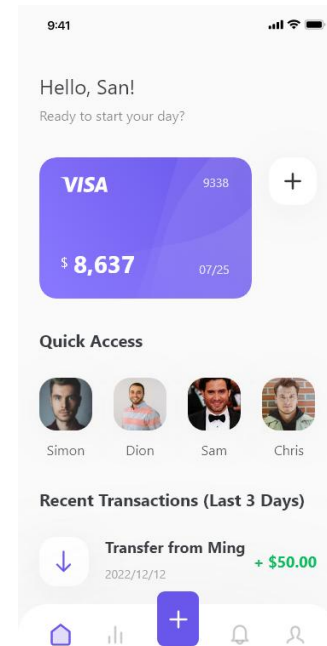
5.Home Page

5.1 Elements include:

- 1) **【Hello, San!】** text;
- 2) **【Ready to start your day?】** text;
- 3) A credit card info section:
 - a) Rounded card;
 - b) Displays the VISA brand;
 - c) Last four digits of the card number;
 - d) Balance;
 - e) Date;
- 4) Add Credit Card button;
- 5) Quick access list;
- 6) Income and expense records for the last three days;
- 7) Bottom navigation bar:
 - a) Including **【Home】** , **【Analytics】** , **【Plus】** , **【Notifications】** , and **【Profile】** .

5.2 Functional requirements:

- 1) Tapping the **【Add Credit Card】** button navigates to the card wallet page;
- 2) The credit card information is sourced from transaction_history.json;
- 3) Quick access list:
 - a) Includes a title and a list of user items;
 - b) At least 5 users are displayed horizontally; the list can be scrolled left and right;
 - c) Each user item shows an avatar and username;
 - d) Long-pressing a user's avatar displays a trash can icon;
 - e) Tapping the trash can icon deletes that user;
 - f) Data is sourced from quick.json;
- 4) The recent three-day income/expense records display at least 10 entries. Each entry shows a title, icon/image, date, and amount;
- 5) The bottom navigation bar:
 - a) The **【Home】** tab is selected by default; tapping it navigates to the home page;
 - b) Tapping the **【Plus】** button navigates to the transfer page;
 - c) Tapping the **【Notifications】** button navigates to the notifications page;
 - d) Tapping the **【Profile】** button navigates to the profile page.



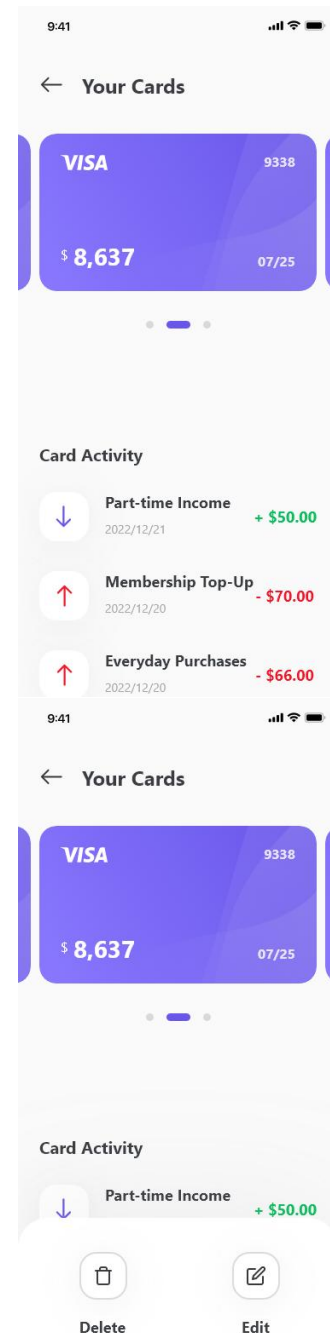
6. Card Wallet Page

6.1 Elements include:

- 1) **【Your Cards】** heading;
- 2) Back button;
- 3) Card carousel:
 - a) VISA logo;
 - b) Last four digits of the card number;
 - c) Balance;
 - d) Date;
 - e) Carousel indicator dots;
- 4) A list displaying transaction records for the cards;

6.2 Functional requirements:

- 1) Card carousel:
 - a) At least 3 cards are displayed;
 - b) Users can swipe left/right to switch between cards;
 - c) The carousel indicator dots update in sync with the active card;
- 2) Card transaction records:
 - a) Each card has its corresponding list of transaction data;
 - b) Data is sourced from transaction_history.json;
 - c) At least 5 records are displayed per card;
- 3) Tapping on a credit card opens a bottom sheet dialog displaying **【Delete】** and **【Edit】** options.



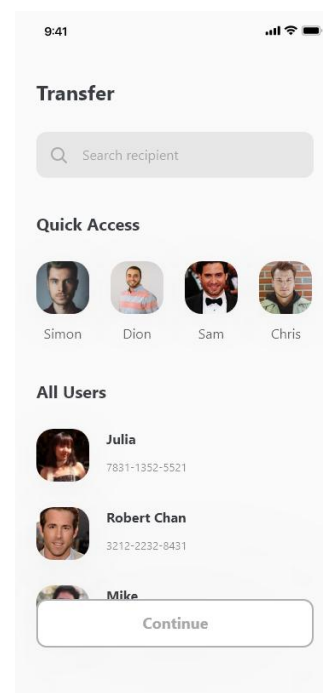
7. Transfer Page

7.1 Elements include:

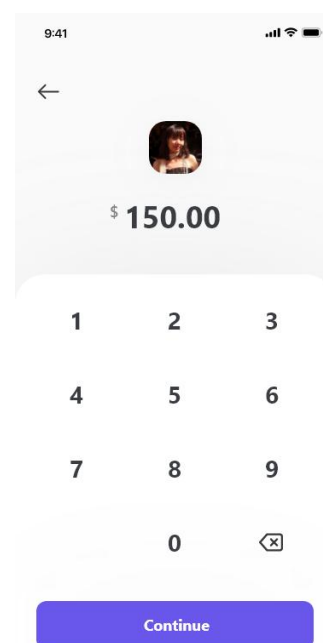
- 1) Transfer heading;
- 2) Search bar;
- 3) Quick Access bar;
- 4) All Users bar;
- 5) Continue button;

7.2 Functional requirements:

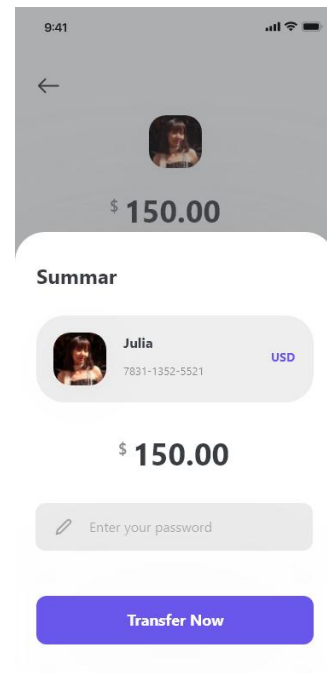
- 1) Access this page by tapping the **【Plus】** button in the bottom navigation bar;
- 2) Quick Access bar:
 - a) Supports horizontal swiping left/right;
 - b) Displays at least 5 entries;
 - c) Each entry shows an avatar and name;
- 3) All Users bar:
 - a) Supports vertical scrolling;
 - b) Displays at least 10 users;
 - c) Each user entry displays an avatar, name, and phone number;
 - d) Data is sourced from user_directory.json;
- 4) Select a user, then tap the Continue button to navigate to the payment page;
- 5) Payment page includes:
 - a) The selected user's avatar;
 - b) Back button;
 - c) Numeric keypad with buttons 1-9;
 - d) Delete button;
 - e) Continue button;
- 6) Enter the amount using the numeric keypad, then tap Continue to display the password dialog;
- 7) Tap "Transfer Now" to display the transfer success dialog;
- 8) Tap **【Back to Home】** to return to the home page.



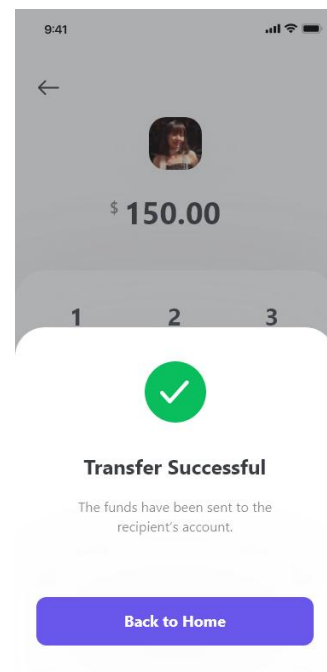
Transfer Page



Payment Page



Payment Page – Password Dialog



Payment Page – Transfer Successful

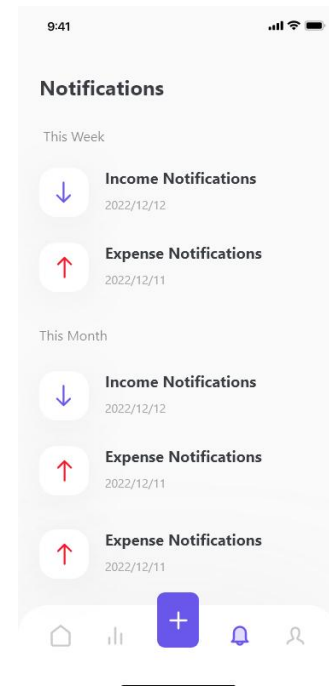
8. Notifications Page

8.1 Elements include:

- 1) Notifications heading;
- 2) Notification records;
- 3) Bottom navigation bar:
 - a) Including **【Home】** , **【Analytics】** , **【Plus】** , **【Notifications】** , and **【Profile】** ;

8.2 Functional requirements:

- 1) Access this page by tapping the **【Notifications】** button in the bottom navigation bar;
- 2) Notification records:
 - a) Display income notifications;
 - b) Display expense notifications;
 - c) Each notification entry includes an avatar, title, and date;
 - d) Data is sourced from notifications.json;
- 3) Notifications are categorized into "This Week" and "This Month";
- 4) At least 5 notifications are displayed under "This Week", and at least 8 under "This Month".



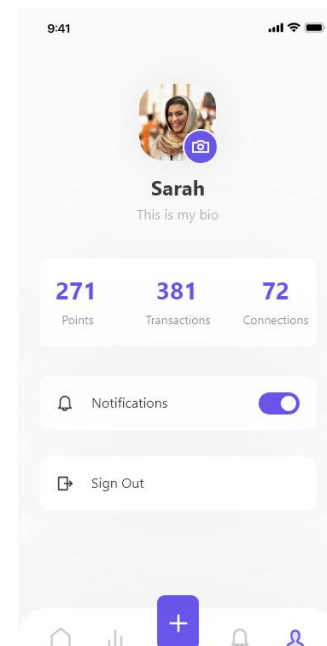
9. Profile Page

9.1 Elements include:

- 1) User avatar;
- 2) Camera button;
- 3) Reward Points;
- 4) Transactions;
- 5) Connections;
- 6) Notification toggle;
- 7) **【Sign Out】** button;
- 8) Bottom navigation bar:
 - a) Including **【Home】** , **【Analytics】** , **【Plus】** , **【Notifications】** , and **【Profile】** ;

9.2 Functional requirements:

- 1) Access this page by tapping the **【Profile】** button in the bottom navigation bar;
- 2) The notification toggle can be tapped to turn notifications on/off;
- 3) Tapping the **【Sign Out】** button displays a dialog asking, "Exit the app?". Tapping **【Yes】** closes the app, and tapping **【No】** dismisses the dialog.



Instructions to the Competitor

1. Please create the project with correct package name (Android) / Organization Identifier (iOS). Naming convention: edu.ws2026.aXX;
2. You should save the project in the folder: XX_Module_A;
3. You should rename the generated apk (Android) / app (iOS) file as XX_Module_A.apk (Android) / XX_Module_A.app (iOS), and save it in the root of the XX_Module_A folder;
4. The whole XX_Module_A folder should be pushed to the provided remote Git repository;
5. Note: XX refers to your workstation code.